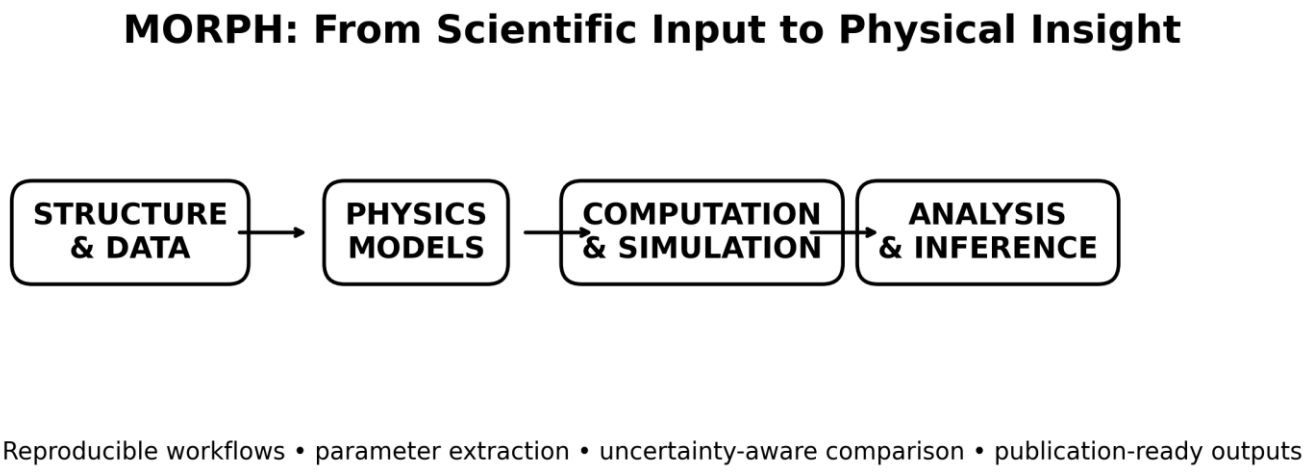
MORPH

Materials-Oriented Robotic Physics Helper

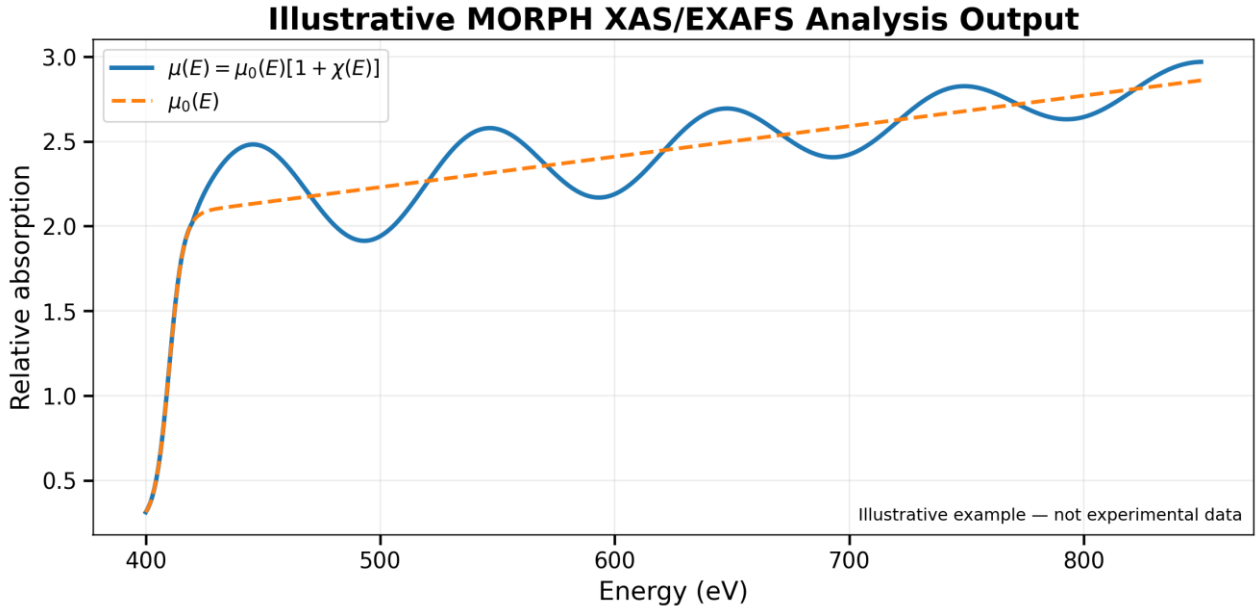
A computational scientific platform that turns materials data and physics models into reproducible, interpretable results.



What MORPH delivers

MORPH is designed as a scalable scientific-workflow framework for materials physics and spectroscopy. It links structure generation, physics-based simulation, data reduction, visualization, parameter extraction, and comparison with theory or experiment. Rather than treating each calculation as an isolated task, MORPH organizes the complete scientific chain so that assumptions, inputs, outputs, and derived quantities remain traceable.

Core applications: XPS, XAS/XANES/EXAFS, FEFF workflows, electronic and structural modeling, spectral decomposition, bond-length and coordination analysis, multivariate/PCA analysis, uncertainty studies, and automated figure/report generation.



MORPH transforms computational physics from a sequence of manual calculations into a documented scientific workflow — from atomic structure to measurable observables and physical interpretation.